

Project Initialization and Planning Phase

Date	28 June 2026
Team ID	xxxxxx
Project Title	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to improve Human Development Index (HDI) analysis and prediction using machine learning techniques, enhancing accuracy and efficiency in assessing a country's development status. The system addresses the challenges of manual analysis of socio-economic indicators by providing automated predictions and insightful visualizations. This solution supports researchers, policymakers, and analysts in making informed decisions. Key features include a machine learning-based HDI prediction model, interactive data visualization, and a web-based application for real-time prediction and analysis.

Project Overview	
Objective	To develop a machine learning-based system that predicts the Human Development Index (HDI) using socio-economic indicators such as life expectancy, education, and income.
Scope	The project focuses on analyzing HDI-related data, training a machine learning model, and deploying a web application for predicting HDI scores and generating visual reports.
Problem Statement	
Description	Manual analysis of multiple HDI indicators is complex and time-consuming, making it difficult to assess and compare development levels accurately.
Impact	Solving this problem will improve decision-making, enable faster analysis, and provide accurate predictions for researchers and policymakers.
Proposed Solution	
Approach	Use machine learning techniques to analyze HDI indicators and predict the Human Development Index through a web-based application.
Key Features	- Machine learning-based HDI prediction. - Interactive data visualization. - Web application for user input and prediction. - Report generation and comparative analysis.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard CPU or T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle HDI Dataset, CSV